ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 162	Let breadth $= b$.
2	**(a)** 16	Perimeters are equal.
3	**(d)** 120	Half of the rectangle.
4	**(b)** 336	Outer minus inner.
5	**(c)** 14	Perimeter does not change.
6	**(d)** 5	Area = cost ÷ rate; breadth = area ÷ length.
7	**(b)** 64	A square has the greatest area for a given perimeter.
8	**(b)** 42	Add the three sides.
9	**(a)** Both A and R are true and R is the correct explanation of A.	2 x (12 + 5) = 34.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	7 x 7 = 49 and area is in square units, but R does not explain the value 49.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$l + b = 30$; ratio parts $= 5$ (1); Length $= 18$ cm, breadth $= 12$ cm (1).	1+1
12	Area $= 1200$ m^2, perimeter $= 140$ m; Levelling cost $= 6000$ (1); Fencing cost $= 1680$; total $= 7680$ (1).	1+1
13	Room area $= 48$ m^2; Carpet area $= 20$ m^2 (1); Uncovered $= 28$ m^2 (1).	1+1
14	$l + b = 32$; ratio parts $= 8$ (1); Length $= 20$ cm, breadth $= 12$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Perimeter $= 130$ m; Wire needed $= 3 \times 130 = 390$ m (1); Fencing cost $= 4680$ (1); Ploughing cost $= 4000$ (1).	1+1+1
16	Square perimeter $= 4 \times 12 = 48$ cm; rectangle perimeter $= 2(14 + 10) = 48$ cm; Square area $= 144$ cm^2 (1); Rectangle area $= 140$ cm^2 (1); Square is larger by 4 cm^2 (1).	1+1+1
17	Floor in cm: 600 by 400; Tile area 600 cm^2 (1); Floor area 240000 cm^2 (1); Tiles $= 400$ (1).	1+1+1
	OR: Perimeter $= 160$ m; Wire needed $= 2 \times 160 = 320$ m (1); Fencing cost $= 4800$ (1); Ploughing cost $= 7500$ (1).	1+1+1
18	Square perimeter $= 4 \times 15 = 60$ cm; rectangle perimeter $= 2(18 + 12) = 60$ cm; Square area $= 225$ cm^2 (1); Rectangle area $= 216$ cm^2 (1); Square is larger by 9 cm^2 (1).	1+1+1
	OR: Floor in cm: 800 by 500; Tile area 1000 cm^2 (1); Floor area 400000 cm^2 (1); Tiles $= 400$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2 \times (40 + 25) = 130$ m (1). (ii) $40 \times 25 = 1000$ m^2 (1). (iii) $130 \times 20 = ₹2600$ (1). (iv) $1000 \div 50 = 20$ parts, so $20 \times 2 = 40$ kg (2).	1+1+1+2
20	Rectangle area $= 240$ cm^2 (1); Triangle area $= \frac{1}{2} \times 24 \times 6 = 72$ cm^2 (1); Remaining $= 168$ cm^2 (1); Fraction left $= \frac{168}{240}$ (2).	1+1+1+2
	OR: Path parallel to length: $80 \times 4 = 320$ m^2 (1); Path parallel to breadth: $50 \times 4 = 200$ m^2 (1); Common square $= 4 \times 4 = 16$ m^2 (1); Area of paths $= 504$ m^2 (1); Rest of the park $= 3496$ m^2 (1).	1+1+1+1+1

Mistakes to avoid:

- Using area units (cm^2) for perimeter or linear units (cm) for area.
- Forgetting the $\frac{1}{2}$ in the area of a triangle.
- Using the sloping side instead of the perpendicular height in the triangle area.
- Writing $P = l + b$ for a rectangle (it needs $2(l + b)$) or $A = 2(l + b)$.
- Mixing units: multiplying metres by centimetres without converting first.
- Using area when the question is about fencing (perimeter) and the reverse for paving or tiling.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 20	Perimeters are equal.
2	**(c)** 240	Half of the rectangle.
3	**(a)** 584	Outer minus inner.
4	**(b)** 18	Perimeter does not change.
5	**(b)** 7	Area = cost ÷ rate; breadth = area ÷ length.
6	**(c)** 100	A square has the greatest area for a given perimeter.
7	**(a)** 68	Add the three sides.
8	**(c)** 124	Outer side $= a + 2$.
9	**(c)** A is true but R is false.	Breadth = 48 / 8 = 6, so A is true. Breadth = area / length, so R is false.
10	**(d)** A is false but R is true.	A square of perimeter 16 cm has area 16 cm^2 but a 6 cm by 2 cm rectangle with perimeter 16 cm has area 12 cm^2. A is false, R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area $= 1000$ m^2, perimeter $= 140$ m; Levelling cost $= 4000$ (1); Fencing cost $= 2100$; total $= 6100$ (1).	1+1
12	Room area $= 63$ m^2; Carpet area $= 30$ m^2 (1); Uncovered $= 33$ m^2 (1).	1+1
13	$l + b = 49$; ratio parts $= 7$ (1); Length $= 28$ cm, breadth $= 21$ cm (1).	1+1
14	Area $= 2700$ m^2, perimeter $= 210$ m; Levelling cost $= 8100$ (1); Fencing cost $= 2100$; total $= 10200$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Perimeter $= 190$ m; Wire needed $= 3 \times 190 = 570$ m (1); Fencing cost $= 5700$ (1); Ploughing cost $= 6300$ (1).	1+1+1
16	Square perimeter $= 4 \times 20 = 80$ cm; rectangle perimeter $= 2(24 + 16) = 80$ cm; Square area $= 400$ cm^2 (1); Rectangle area $= 384$ cm^2 (1); Square is larger by 16 cm^2 (1).	1+1+1
17	Floor in cm: 900 by 600; Tile area 450 cm^2 (1); Floor area 540000 cm^2 (1); Tiles $= 1200$ (1).	1+1+1
	OR: Perimeter $= 130$ m; Wire needed $= 4 \times 130 = 520$ m (1); Fencing cost $= 4160$ (1); Ploughing cost $= 5400$ (1).	1+1+1
18	Square perimeter $= 4 \times 18 = 72$ cm; rectangle perimeter $= 2(23 + 13) = 72$ cm; Square area $= 324$ cm^2 (1); Rectangle area $= 299$ cm^2 (1); Square is larger by 25 cm^2 (1).	1+1+1
	OR: Floor in cm: 1000 by 400; Tile area 1000 cm^2 (1); Floor area 400000 cm^2 (1); Tiles $= 400$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $12 \times 9 = 108$ m^2 (1). (ii) $3 \times 3 = 9$ m^2 (1). (iii) $108 \div 9 = 12$ slabs (1). (iv) Total cost $= 12 \times 450 = ₹5400$ (1); cost per m^2 $= 450 \div 9 = ₹50$ (1).	1+1+1+2
20	Rectangle area $= 144$ cm^2 (1); Triangle area $= \frac{1}{2} \times 16 \times 6 = 48$ cm^2 (1); Remaining $= 96$ cm^2 (1); Fraction left $= \frac{96}{144}$ (2).	1+1+1+2
	OR: Path parallel to length: $90 \times 5 = 450$ m^2 (1); Path parallel to breadth: $60 \times 5 = 300$ m^2 (1); Common square $= 5 \times 5 = 25$ m^2 (1); Area of paths $= 725$ m^2 (1); Rest of the park $= 4675$ m^2 (1).	1+1+1+1+1

Mistakes to avoid:

- Using area units (cm^2) for perimeter or linear units (cm) for area.
- Forgetting the $\frac{1}{2}$ in the area of a triangle.
- Using the sloping side instead of the perpendicular height in the triangle area.
- Writing $P = l + b$ for a rectangle (it needs $2(l + b)$) or $A = 2(l + b)$.
- Mixing units: multiplying metres by centimetres without converting first.
- Using area when the question is about fencing (perimeter) and the reverse for paving or tiling.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 252	Half of the rectangle.
2	**(d)** 252	Outer minus inner.
3	**(c)** 21	Perimeter does not change.
4	**(a)** 6	Area = cost ÷ rate; breadth = area ÷ length.
5	**(a)** 144	A square has the greatest area for a given perimeter.
6	**(a)** 84	Add the three sides.
7	**(c)** 363	Let breadth = b.
8	**(d)** 36	Perimeters are equal.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Half of 10 x 6 = 30.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Perimeter is the length of the boundary.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Room area = 60 m^2; Carpet area = 28 m^2 (1); Uncovered = 32 m^2 (1).	1+1
12	$l + b = 36$; ratio parts = 12 (1); Length = 21 cm, breadth = 15 cm (1).	1+1
13	Area = 875 m^2, perimeter = 120 m; Levelling cost = 5250 (1); Fencing cost = 960; total = 6210 (1).	1+1
14	Room area = 108 m^2; Carpet area = 48 m^2 (1); Uncovered = 60 m^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Perimeter = 220 m; Wire needed = $2 \times 220 = 440$ m (1); Fencing cost = 8800 (1); Ploughing cost = 5600 (1).	1+1+1
16	Square perimeter = $4 \times 25 = 100$ cm; rectangle perimeter = $2(31 + 19) = 100$ cm; Square area = 625 cm^2 (1); Rectangle area = 589 cm^2 (1); Square is larger by 36 cm^2 (1).	1+1+1
17	Floor in cm: 1200 by 500; Tile area 1200 cm^2 (1); Floor area 600000 cm^2 (1); Tiles = 500 (1).	1+1+1
	OR: Perimeter = 170 m; Wire needed = $3 \times 170 = 510$ m (1); Fencing cost = 7140 (1); Ploughing cost = 6600 (1).	1+1+1
18	Square perimeter = $4 \times 16 = 64$ cm; rectangle perimeter = $2(23 + 9) = 64$ cm; Square area = 256 cm^2 (1); Rectangle area = 207 cm^2 (1); Square is larger by 49 cm^2 (1).	1+1+1
	OR: Floor in cm: 700 by 600; Tile area 700 cm^2 (1); Floor area 420000 cm^2 (1); Tiles = 600 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $12 + 9 + 15 = 36$ m (1). (ii) $\frac{1}{2} \times 12 \times 9 = 54$ m^2 (1). (iii) $12 \times 9 = 108$ m^2 (1). (iv) Cost = $36 \times 30 = ₹1080$ (1); the triangle is half of the rectangle ($54 = 108 \div 2$) (1).	1+1+1+2
20	Rectangle area = 420 cm^2 (1); Triangle area = $\frac{1}{2} \times 28 \times 10 = 140$ cm^2 (1); Remaining = 280 cm^2 (1); Fraction left = $\frac{280}{420}$ (2).	1+1+1+2
	OR: Path parallel to length: $50 \times 2 = 100$ m^2 (1); Path parallel to breadth: $30 \times 2 = 60$ m^2 (1); Common square = $2 \times 2 = 4$ m^2 (1); Area of paths = 156 m^2 (1); Rest of the park = 1344 m^2 (1).	1+1+1+1+1

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- Using the sloping side instead of the perpendicular height in the triangle area.
- Writing $P = l + b$ for a rectangle (it needs $2(l + b)$) or $A = 2(l + b)$.
- Mixing units: multiplying metres by centimetres without converting first.
- Using area when the question is about fencing (perimeter) and the reverse for paving or tiling.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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